**Biology Class 11 National Book Foundation FBISE Quality Notes**
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# Chapter 10: Animalia

- Name **Animalia** from Latin **anima** = breath or soul. Animals are **eukaryotic, multicellular, ingestive heterotrophs that lack cell walls**. They live in almost all habitats: free living motile, sessile or parasitic; size from microscopic worms to blue whales.

## 10.1 Criteria for animal classification

- Animals are classified by (a) presence or absence of tissue, (b) number of tissue layers, (c) body symmetry, (d) type of body cavity.

| Criterion | Groups |
|---|---|
| Organization of tissues | **Parazoa** (no tissues, asymmetrical, division of labour: sponges) and **Eumetazoa** (cells grouped into tissues, organs, organ systems). |
| Number of tissue layers | **Diploblastic** (ectoderm + endoderm with jelly-like mesoglea; one gastrovascular cavity with one opening; nerve net: Cnidaria). **Triploblastic** (ectoderm, mesoderm, endoderm). |
| Body symmetry | **Radiata** (radial: spokes from central axis; adaptation for sessile life; jellyfish, sea anemone). **Bilateria** (bilateral: mirror halves, head, tail, dorsal and ventral surfaces; adaptation to motility; all triploblastic). Echinoderm larvae are bilateral, adults secondarily radial. |
| Body cavity (coelom) | **Acoelomate** (no cavity; mesoderm solid: flatworms), **pseudocoelomate** (cavity between mesoderm and endoderm, false coelom: roundworms), **coelomate** (cavity within mesoderm, completely lined by mesoderm, filled with coelomic fluid: annelids to chordates). |

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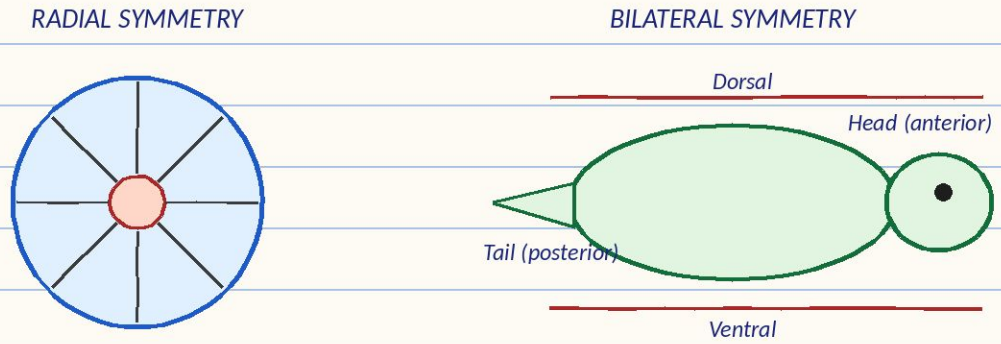



Fig: Radial and bilateral symmetry

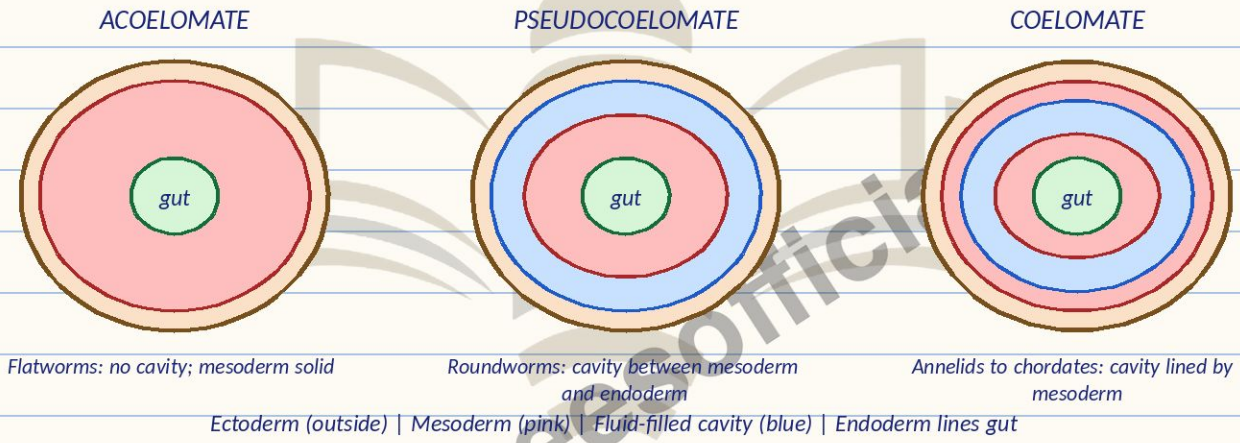
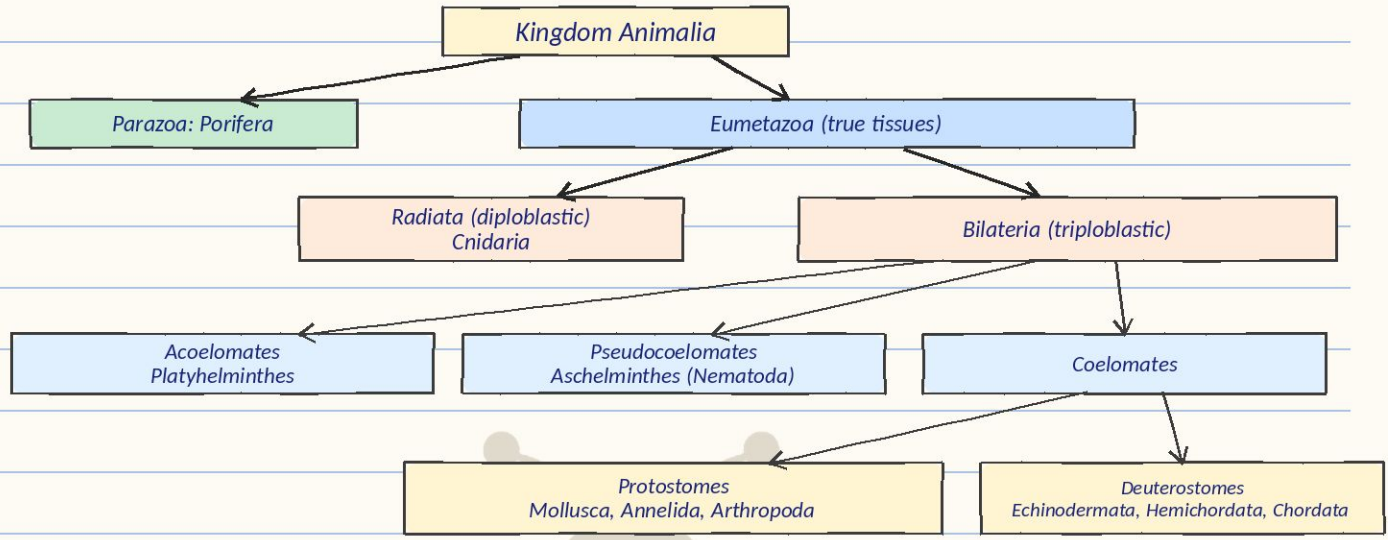



Fig: Acoelomate, pseudocoelomate and coelomate body plans (T.S.)

- ***Coelomates*** *split into* ***protostomes*** *(Gk first mouth: the blastopore becomes the mouth: molluscs, annelids, arthropods) and* ***deuterostomes*** *(second mouth: blastopore becomes the anus, mouth forms later: echinoderms and chordates). Blastopore = opening formed when a group of cells of the blastula moves inward.*

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*Protostome: blastopore becomes mouth; Deuterostome: blastopore becomes anus*

*Fig: Basis of animal classification (tissues, symmetry, coelom, blastopore)*

## 10.2 Diversity in animals (SLO 1)

- *Animals are divided into **invertebrates** (no backbone; 95% of known species; eight major phyla) and **vertebrates** (backbone).*

## Invertebrates: eight phyla

### 1. Phylum Porifera (pore bearer): sponges

- *Sessile (larvae motile); aquatic, mostly marine; mm to over 1 m; multicellular, no tissues or organs, **no symmetry**. Outer **pinacoderm** (flat **pinacocytes**); inner **choanoderm** (flagellated **choanocytes**, collar cells); middle **mesenchyme** with amoeboid cells. Many pores **ostia**; single cavity **spongocoel**; water in through ostia, out by large **osculum**. Feed on food in water currents. Skeleton of **spicules** (calcareous or siliceous); bath sponge has **spongin fibre**.*

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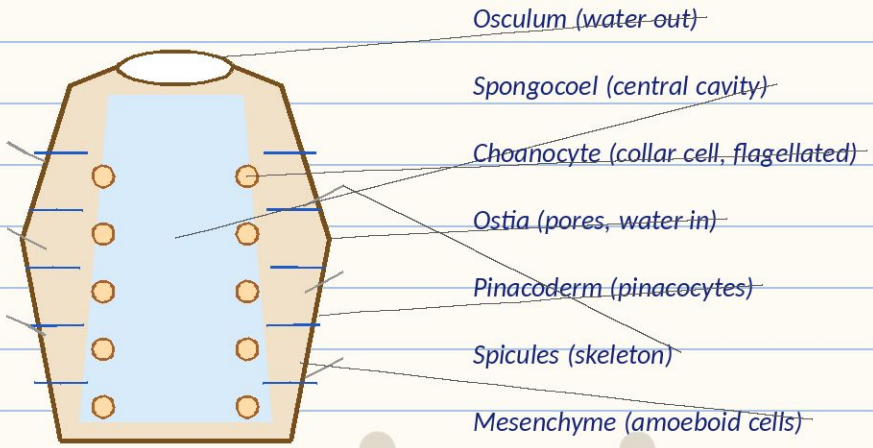
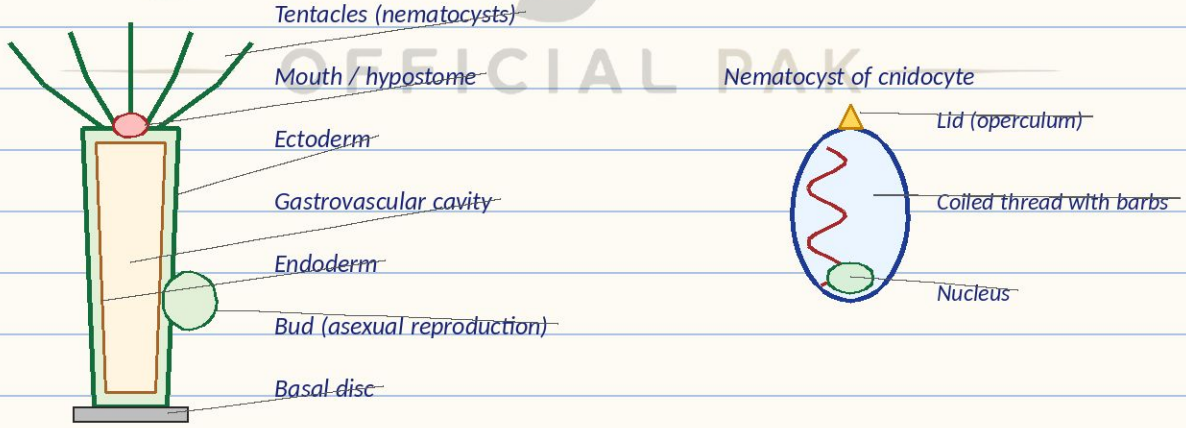



Fig: Structure of a sponge (Sycon type)

- Asexual: budding, **gemmules**. Sexual: eggs and sperm; separate sexes or hermaphrodite; free-swimming ciliated larva. Remarkable regeneration. Evolved from choanoflagellates. Examples: **Spongilla, Sycon**.
- **Importance:** skeletons for bathing and washing; absorb water, used in surgery to absorb blood and fluid; sound absorption in buildings.

## 2. Phylum Cnidaria (stinging cell organisms)

- Aquatic, mainly marine (Hydra freshwater); sessile (Hydra) or free (jellyfish) or colonial (Obelia); microscopic to 2 m. **Radial or biradial symmetry**; **diploblastic** with ectoderm, endoderm and **mesoglea**; sac-like **gastrovascular cavity** with a single opening (mouth) ringed by tentacles; **nematocysts** (stinging organelles in **cnidocytes**); nerve net; asexual by budding, sexual by gametes.




Polyp (sessile, feeding, e.g. Hydra, Obelia) vs Medusa (free swimming, reproductive, e.g. jellyfish)

Fig: Hydra (polyp) and a nematocyst

- Colonies of **zooids**: **polyps** (feeding, sedentary, mouth and tentacles) and

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***medusae*** *(free swimming, reproductive).* ***Polymorphism*** *= division of labour among zooids (Obelia dimorphic; colony has* ***hydranths***, ***blastostyles*** *and medusae).*

- ***Corals:*** *ectodermal cells take lime from sea water and make* $CaCO_3$ *exoskeleton that forms* ***coral reefs*** *and islands (Florida, West Indies, East Africa, Australia, Coral Sea).* ***Importance:*** *reefs protect shores from erosion by tidal waves; jewellery, aquaria; jellyfish and sea anemone have poisonous stings.*

### 3. Phylum Platyhelminthes (flatworms)

- *Free living (****Planaria / Dugesia****) or parasitic (****Taenia*** *tapeworm,* ***Fasciola*** *liver fluke); freshwater, marine, gut, liver. Soft,* ***dorsiventrally flattened***; ***triploblastic, bilateral***; ***acoelomate*** *(mesenchyme fills spaces); eyespots in some; free-living forms move by cilia on underside. Asexual by* ***fission***; *most* ***monoecious***; *complex reproductive system; internal fertilization.*
- ***Parasitic adaptations:*** *hooks and suckers (scolex of tapeworm), no digestive system in tapeworm (absorb food), thick cuticle, high reproductive capacity.* ***Importance:*** *tapeworm, liver fluke and blood fluke of cattle are harmful.*

### 4. Phylum Aschelminthes (Nematoda): roundworms

- *Free living or parasites in soil, roots, human and animal intestine and muscles; mostly under 5 cm (some over 1 m).* ***Bilateral, triploblastic****, cylindrical body tapering at both ends; muscle layer divided into four longitudinal quadrants (2 dorsolateral, 2 ventrolateral);* ***pseudocoelom***; *mostly* ***dioecious***; *internal fertilization.*
- ***Importance: Ascaris lumbricoides*** *(intestinal parasite); pinworms* ***Enterobius vermicularis*** *in human caecum, colon and appendix cause insomnia and loss of appetite; Cyclops carries larvae of Guinea worm.*

### 5. Phylum Mollusca (soft bodied)

- *Free living or sessile; freshwater, marine, moist land.* ***Bilateral, triploblastic, coelomate****, soft, unsegmented. Body:* ***head, ventral muscular foot, dorsal visceral mass****, covered by* ***mantle*** *which secretes the* ***shell*** *(external in snail, internal in cuttlefish, absent in octopus). Mantle cavity: excretion, gas*

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*exchange, waste and reproductive products. **Radula** (tongue-like) in snail and cuttlefish. Sexes separate (Unio) or united (Helix). Development through **trochophore larva**.*

- *Examples: snail, slug, oyster, freshwater mussel, octopus. **Importance:** mussel shells for buttons and ornaments; pearl oysters give pearls; clams, oysters, mussels as food; slugs injure gardens; **Teredo** (shipworm) damages wood of ships.*

## *6. Phylum Annelida (segmented worms)*

- *Free living (earthworm), ectoparasite (leech), soil, freshwater, marine (**Nereis**). Body **metamerically segmented**; **true coelom** divided into compartments, fluid mixes between chambers and works as a **hydrostatic skeleton**; circular and longitudinal muscles; locomotion by chitinous **setae (chaetae)**, **parapodia** in Nereis; closed circulatory system; ventral nerve cord with ganglia; hermaphrodite (earthworm, leech) or separate sexes (Nereis); regeneration common; direct development or **trochophore larva**.*
- *Examples: **Nereis**, **Pheretima** (earthworm), **Hirudo** (leech). **Importance:** polychaetes are food for fish; earthworms improve soil; leech is an ectoparasite of man and cattle.*

## *7. Phylum Arthropoda (jointed legs)*

- *Free living or parasites in all habitats; segmented body with segments modified, specialized and fused; **bilateral**; head, thorax, abdomen; covered by flexible **chitinous exoskeleton** with joints; several pairs of jointed appendages; coelom reduced = **haemocoel** (connected with blood vascular system); complete alimentary canal; muscles attached to exoskeleton; sexes separate; mostly internal fertilization; **metamorphosis**.*
- *Success of insects: ability to fly, protective mechanisms (cutin cover, mimicry, protective colouration) and young and adult use different food and habitats. Examples: prawn, Cyclops, butterfly, spider.*
- ***Importance:** lobsters, crayfish, prawns eaten; honey and wax from honeybee; silk from silkworm; pollination; laboratory animals (Drosophila, cockroach,*

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*grasshopper); **harmful:** destroy crops, fruit trees and timber; spread malaria and dengue; bees and wasps sting; mosquitoes, lice and fleas suck blood; scorpions and a few spiders poisonous; Cyclops host of Guinea worm.*

## *8. Phylum Echinodermata (spiny skinned)*

- *Exclusively marine; free living or attached; **triploblastic coelomates**, **radial symmetry in adult** (bilateral larvae). **Endoskeleton** of spine-bearing calcareous plates. Mouth on oral side, anus on aboral. **Water vascular system** (water-filled canals from coelom) for locomotion, attachment, gas exchange and feeding; **tube feet**; nerve ring and radial nerves. Sexes separate; external fertilization; larvae (bipinnaria, brachiolaria) bilateral; **autotomy and regeneration**.*
- *Examples: starfish, sea cucumber, sea lily, brittle star, sea urchin. **Importance:** used as food; dried skeletons as fertilizer (calcium, nitrogen); starfish scavenge but damage oyster beds; sea urchins poisonous.*
- ***Echinoderms resemble chordates:** mesodermal skeleton; deuterostomous; similar early development.*

## *9. Phylum Hemichordata (acorn worms)*

- *Show characters of echinoderms and chordates (deuterostomes); called prochordates. Marine; soft, unsegmented worm-like; three regions: **proboscis, collar, trunk**; bilateral, triploblastic, true coelom; **glomerulus** for excretion (no nephridia); external fertilization. Examples: **Balanoglossus, Saccoglossus**.*

## *10. Phylum Chordata (SLO 2)*

- *Four basic characteristics (at some stage of life):*

1. ***Notochord:** solid unjointed rod in the mid-dorsal line between gut and nerve cord, outside the coelom; axial endoskeleton for support and muscle attachment; in higher chordates partly or wholly replaced by **vertebral column**.*
2. ***Dorsal hollow nerve cord:** single tubular, fluid filled, non-ganglionated, above the notochord.*

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3. ***Gill slits (pharyngeal pouches):*** *paired perforations on sides of pharynx.*
4. ***Post-anal tail:*** *extends beyond the anus; at least in embryo (regresses to tail bone in humans).*




*Four features of every chordate (at some stage)*

*Fig: Main features of chordates (generalized embryo)*

- *Chordates have endoskeleton, closed circulatory system, paired appendages, efficient respiration and excretion, and high cephalisation.*

## Classification of chordates

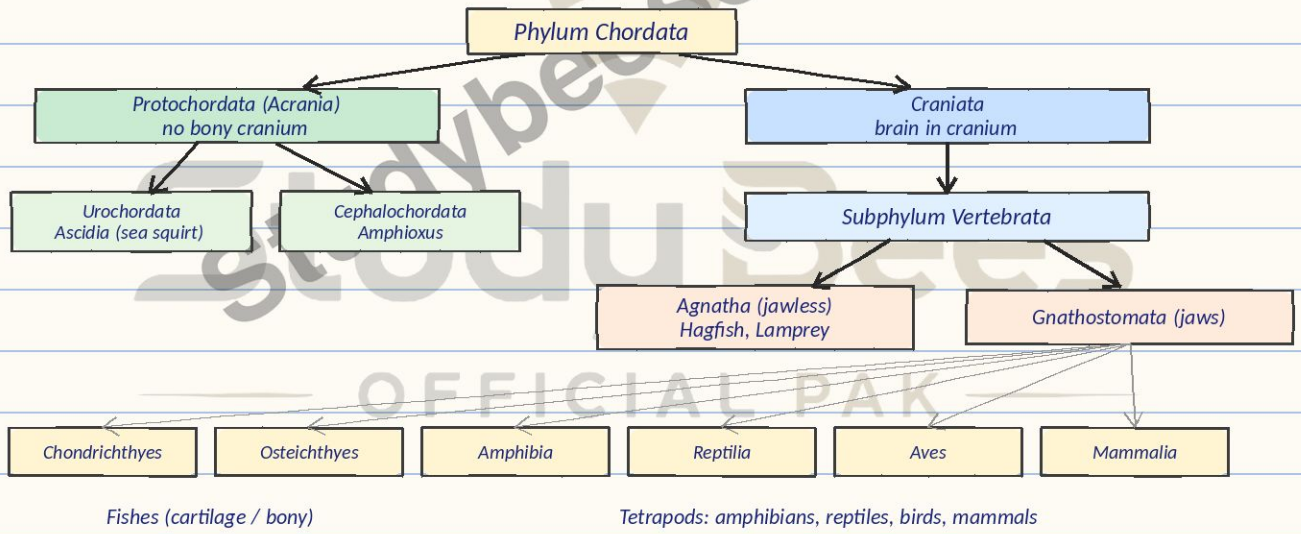



*Fig: Classification of phylum Chordata*

- ***Protochordata (Acrania):*** *brain not in bony case.* ***Urochordata*** *(notochord only in larval tail, lost at metamorphosis:* ***Ascidia****) and* ***Cephalochordata*** *(fish-like, no head, notochord along whole body:* ***Amphioxus****).* ***Craniata:*** *brain in cranium, notochord replaced by vertebral column; subphylum* ***Vertebrata****.*
- *Vertebrata: super class* ***Agnatha*** *(jawless; Myxini and Cephalaspidomorphi;*

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*slender eel-like, naked skin; external fertilization; hagfish, lamprey) and super class* ***Gnathostomata*** *(jaws evolved from the first gill arches). Six classes: Chondrichthyes, Osteichthyes, Amphibia, Reptilia, Aves, Mammalia.*

## *Classes of vertebrates (SLO 3, 4)*

| ***Class*** | ***Features and evolutionary adaptations*** |
|---|---|
| *Chondrichthyes (cartilaginous fishes)* | *Laterally compressed, spindle-shaped body; ventral mouth; olfactory sacs not connected to mouth cavity. Cartilage skeleton; examples dogfish shark (Squalus), skate (Raja).* |
| *Osteichthyes (bony fishes)* | *Streamlined spindle body for active swimming; endoskeleton partly or wholly bony; numerous vertebrae; pelvic girdle often absent; notochord persists greatly reduced. Labeo rohita, sea horse, Latimeria (lobe-finned living fossil).* |
| *Amphibia* | *Body of head and trunk; two pairs of pentadactyl limbs (or none: caecilians); webbed feet; smooth moist glandular, highly vascular skin without scales; gills in larva, lungs and skin in adult;* ***three-chambered heart*** *(2 atria + 1 ventricle), double circulation; separate sexes; oviparous; metamorphosis; anamniotes;* ***poikilothermic****; hibernate. Frogs, toads, salamanders, caecilians, mud puppy.* |
| *Reptilia* | *Two pairs of pentadactyl limbs (5 digits); rough dry cornified skin (land adaptation); heart incompletely four-chambered (crocodile completely); poikilothermic; internal fertilization; large amniotic, yolky eggs with leathery or limy shell; embryonic membranes amnion, allantois, chorion. Tortoise, lizard, snake, crocodile, alligator.* |

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| Class | Features and evolutionary adaptations |
|---|---|
| Aves (birds) | Streamlined boat-shaped body: head, long neck, trunk, short tail; forelimbs as wings, strong hind limbs with four toes and claws; **feathers** and foot scales; light skeleton with air spaces; horny beak, no teeth; **four-chambered heart**, only right aorta; **endothermic**; lungs only with thin-walled **air sacs**; **syrinx** voice box; **gizzard**; kidneys, no urinary bladder, uric acid, cloaca; only one ovary and oviduct; large yolky eggs. Running birds: ostrich, kiwi. |
| Mammalia | Head, neck, trunk, tail; two pairs pentadactyl limbs; skin glandular, with **hair**; **diaphragm** divides thoracic and abdominal cavities (only mammals); fully ossified skeleton; skull with two occipital condyles; external ear (pinna); three ear bones (malleus, incus, stapes); deciduous and permanent teeth; four-chambered heart, only left aortic arch; non-nucleated RBCs; **endothermic**; larynx and epiglottis; give birth and nurse with **mammary glands**. |

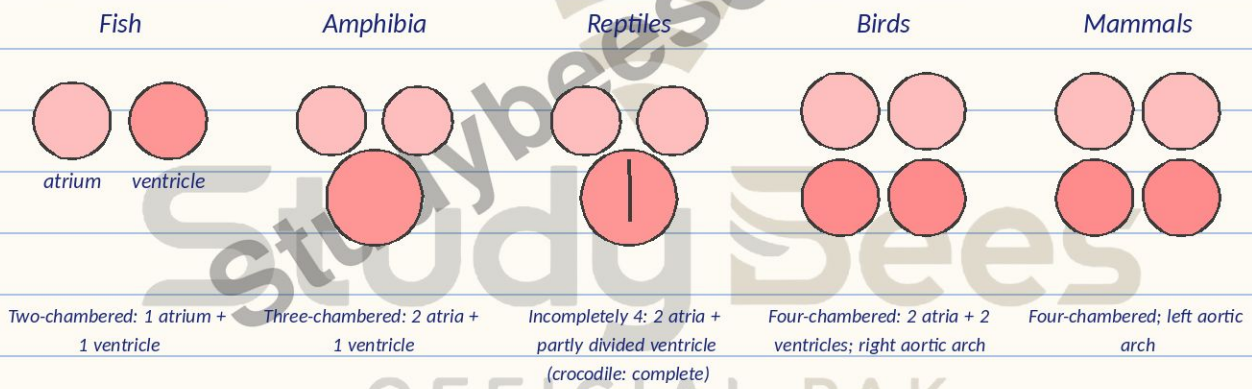



Fig: Heart chambers in vertebrate classes

## Jaws, limbs and amniotes

- ***Jaws evolve:*** *jaws (tooth-bearing bones) evolved from the first pair of gill arches of agnathans.* ***Limbs evolve:*** *lobe-finned fishes of Devonian are ancestral to amphibians, the first tetrapods; limbs support the body on land (air is less buoyant than water); lobe-finned fishes and early amphibians had lungs and internal nares.*

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- ***Amniote egg:*** *internal fertilization and a shelled egg free reproduction from water. Extra-embryonic membranes protect the embryo; the* ***amnion*** *is a fluid filled "private pond". Reptiles, birds and mammals are* ***amniotes****.*
- ***Amphibians:*** *borderline between aquatic and terrestrial animals; depend on water for habitat, reproduction and development, so not a very successful group.*

## Subclasses of Mammalia

| Subclass | Features and examples |
|---|---|
| Prototheria (monotremes) | Egg laying; link between reptiles and mammals; female has mammary glands (mammalian) and cloaca (reptilian); found in Australia. Duckbill platypus (webbed toes, bill), Echidna (spiny ant eater). |
| Metatheria (marsupials) | Pouched mammals; young born immature and reared in the abdominal pouch **marsupium** with nipples; Australia and America. Opossum, kangaroo, koala. |
| Eutheria (placentals) | Young develop inside mother, dependent on **placenta** (exchange between maternal and fetal blood); born at advanced stage; maximum mammalian features; hair modified to scales in pangolin and spines in porcupine. Man, whale, elephant, horse, rat, bat, dolphin, cat, tiger, lion, monkey, gorilla. |

## Land adaptations shared by arthropods and vertebrates

- *Waterproof body covering (chitinous cuticle / keratinized skin) against desiccation; internal respiratory organs (tracheae / lungs); jointed appendages or limbs for support and locomotion; internal fertilization; protected eggs (shells, amnion); efficient excretion (uric acid) and circulatory system.*

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## *Chapter 10: Short Questions with Answers (SLO based)*

***Q1. [SLO 1] What does the term Animalia mean?***

*Ans: It comes from Latin anima = breath or soul; the kingdom includes all animals.*

***Q2. [SLO 1] Write the distinct features of animals.***

*Ans: Eukaryotic, multicellular, ingestive heterotrophs that lack cell walls; motile or sessile.*

***Q3. [SLO 1] Name the four criteria for animal classification.***

*Ans: Presence or absence of tissue, number of tissue layers, body symmetry and type of body cavity.*

***Q4. [SLO 1] Differentiate Parazoa and Eumetazoa.***

*Ans: Parazoa: simplest, no tissue organization, asymmetrical (sponges). Eumetazoa: cells grouped into tissues, organs and organ systems.*

***Q5. [SLO 1] Define diploblastic and triploblastic animals.***

*Ans: Diploblastic: two germ layers (ectoderm, endoderm) with mesoglea (Cnidaria). Triploblastic: three germ layers (ectoderm, mesoderm, endoderm).*

***Q6. [SLO 1] What is mesoglea?***

*Ans: The jelly-like non-living layer between ectoderm and endoderm of diploblastic animals.*

***Q7. [SLO 1] Define radial and bilateral symmetry. Which is an adaptation for sessile life?***

*Ans: Radial: parts radiate from a central axis (adaptation for sessile life). Bilateral: median plane divides body into mirror halves (adaptation to motility).*

***Q8. [SLO 1] Name two phyla with radial symmetry and two with bilateral symmetry.***

*Ans: Radial: Cnidaria and Echinodermata (adults). Bilateral: Platyhelminthes and Annelida (also Arthropoda, Chordata).*

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**Q9. [SLO 1] Define coelom. Differentiate acoelomate and pseudocoelomate.**

*Ans: Coelom: fluid filled cavity between body wall and gut lined by mesoderm. Acoelomate: no cavity (flatworms). Pseudocoelomate: false cavity between mesoderm and endoderm (roundworms).*

**Q10. [SLO 1] Differentiate protostomes and deuterostomes.**

*Ans: Protostomes: blastopore becomes the mouth (molluscs, annelids, arthropods). Deuterostomes: blastopore becomes the anus (echinoderms, chordates).*

**Q11. [SLO 1] What are invertebrates? Name the eight phyla.**

*Ans: Animals without backbone: Porifera, Cnidaria, Platyhelminthes, Aschelminthes, Mollusca, Annelida, Arthropoda, Echinodermata.*

**Q12. [SLO 2] Describe the body layers of sponges.**

*Ans: Outer pinacoderm of pinacocytes, inner choanoderm of flagellated choanocytes and middle mesenchyme with amoeboid cells.*

**Q13. [SLO 2] What are ostia, osculum and spongocoel?**

*Ans: Ostia: pores through which water enters; osculum: large opening for water to leave; spongocoel: the central cavity.*

**Q14. [SLO 2] Differentiate pinacoderm and choanoderm.**

*Ans: Pinacoderm: outer layer of flat pinacocytes. Choanoderm: inner layer of flagellated collar cells (choanocytes).*

**Q15. [SLO 2] What are spicules and spongin?**

*Ans: Spicules: needle-like calcareous or siliceous skeleton; spongin: fibres of the bath sponge.*

**Q16. [SLO 2] From which protists did sponges evolve? How do they reproduce?**

*Ans: Choanoflagellates. Asexually by budding or gemmules; sexually by eggs and sperm.*

**Q17. [SLO 2] List the economic importance of sponges.**

*Ans: Used for bathing and washing, absorbing fluid in surgery and sound absorption in buildings.*

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**Q18. [SLO 2] What are nematocysts?**

*Ans: Stinging organelles formed in cnidocytes on the tentacles and body of cnidarians, used for defence and capturing prey.*

**Q19. [SLO 2] Differentiate spongocoel and gastrovascular cavity.**

*Ans: Spongocoel: central cavity of sponge with many ostia and one osculum. Gastrovascular cavity: sac-like cavity of cnidarians with single opening (mouth).*

**Q20. [SLO 2] What are polyps and medusae? Differentiate them.**

*Ans: Polyp: sedentary feeding zooid with mouth and tentacles. Medusa: free-swimming reproductive zooid.*

**Q21. [SLO 2] What is polymorphism? What are zooids?**

*Ans: Polymorphism: different forms of zooids in a colony showing division of labour. Zooids: the individual units of a cnidarian colony.*

**Q22. [SLO 2] How do corals make reefs?**

*Ans: Ectodermal cells take lime from sea water and form a $CaCO_3$ exoskeleton; the exoskeletons form reefs and islands.*

**Q23. [SLO 2] Give three features of platyhelminthes for parasitic mode of life.**

*Ans: Hooks and suckers (scolex), absorption of food through body (no digestive tract in tapeworm), thick cuticle and high reproductive capacity.*

**Q24. [SLO 2] Why are flatworms called acoelomate?**

*Ans: They have no coelom; the spaces between organs are filled with mesenchyme.*

**Q25. [SLO 2] Give three distinguishing features of Aschelminthes.**

*Ans: Pseudocoelom; cylindrical body tapering at both ends; sexes separate (dioecious); muscles in four longitudinal quadrants.*

**Q26. [SLO 2] Name the human intestinal roundworms.**

*Ans: Ascaris lumbricoides and pinworm Enterobius vermicularis.*

**Q27. [SLO 2] What are mantle and radula?**

*Ans: Mantle: envelope covering the visceral mass that secretes the shell. Radula: tongue-like rasping structure in the mouth cavity of snails and cuttlefish.*

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**Q28. [SLO 2] What is a trochophore larva? Name phyla showing it.**

*Ans: A ciliated free-swimming larva of molluscs and some annelids.*

**Q29. [SLO 2] Why is Teredo harmful?**

*Ans: The shipworm (a mollusc) damages wooden parts of ships.*

**Q30. [SLO 2] How does locomotion take place in annelids?**

*Ans: By circular and longitudinal muscles of the body wall working against the hydrostatic skeleton of coelomic fluid, with chitinous setae and parapodia (Nereis).*

**Q31. [SLO 2] What are parapodia and setae?**

*Ans: Setae: chitinous bristles for locomotion. Parapodia: fleshy lateral outgrowths of the body wall of Nereis.*

**Q32. [SLO 2] Write the economic importance of annelids.**

*Ans: Polychaetes are fish food; earthworms improve soil; leeches are ectoparasites of man and cattle.*

**Q33. [SLO 2] Write five salient features of phylum Arthropoda.**

*Ans: Jointed appendages; chitinous exoskeleton; segmented body (head, thorax, abdomen); haemocoel; metamorphosis; separate sexes.*

**Q34. [SLO 2] What is haemocoel?**

*Ans: The reduced coelom of arthropods, which is connected with the blood vascular system.*

**Q35. [SLO 2] List any four harmful roles of insects.**

*Ans: Destroy crops, fruit trees and timber; spread malaria and dengue; sting (bees, wasps); bite and suck blood (mosquitoes, lice, fleas).*

**Q36. [SLO 2] What is metamorphosis?**

*Ans: A change of form through larval stages to the adult (as in insects and amphibians).*

**Q37. [SLO 2] What is the water vascular system?**

*Ans: A system of water-filled canals derived from the coelom in echinoderms, used for locomotion, attachment, gas exchange and feeding.*

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**Q38. [SLO 2] List the similarities between echinoderms and chordates.**

*Ans: Both have a mesodermal skeleton, are deuterostomous and have similar early development.*

**Q39. [SLO 2] Why are hemichordates called prochordates?**

*Ans: They show characters of both echinoderms and chordates and are closely related to chordates.*

**Q40. [SLO 3] List the four characteristics of chordates.**

*Ans: Notochord, dorsal hollow nerve cord, gill slits and post-anal tail.*

**Q41. [SLO 3] What is the notochord? What replaces it in higher chordates?**

*Ans: A solid unjointed rod along the mid-dorsal line giving support; replaced by the vertebral column.*

**Q42. [SLO 3] Differentiate urochordates and cephalochordates.**

*Ans: Urochordates (Ascidia): notochord only in larval tail. Cephalochordates (Amphioxus): fish-like, no head, notochord along whole body.*

**Q43. [SLO 3] Differentiate Agnatha and Gnathostomata.**

*Ans: Agnatha: jawless fishes (hagfish, lamprey). Gnathostomata: animals with jaws, six classes.*

**Q44. [SLO 4] What is the swim bladder? (as asked in book)**

*Ans: A gas-filled sac of bony fishes that helps them regulate buoyancy and stay at a depth.*

**Q45. [SLO 4] Why are amphibians not a very successful vertebrate group?**

*Ans: They depend on water for habitat, reproduction and development.*

**Q46. [SLO 4] What does the word amphibian mean?**

*Ans: Amphi = both, bios = life: animals living both in water and on land (larva aquatic, adult semi-terrestrial).*

**Q47. [SLO 4] How are amphibians adapted to land and what limits them?**

*Ans: Lungs, pentadactyl limbs and moist skin help on land; but they need moist skin, water for eggs without shells and larva with gills.*

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***Q48. [SLO 4] List adaptations that distinguish reptiles from amphibians.***

*Ans: Dry cornified skin, shelled amniotic egg, internal fertilization, better heart (partly four chambered), no larval stage.*

***Q49. [SLO 4] What is an amniote egg? Which animals are amniotes?***

*Ans: An egg with extraembryonic membranes (amnion, allantois, chorion); reptiles, birds and mammals.*

***Q50. [SLO 4] What is amnion?***

*Ans: The membrane forming a fluid-filled sac that provides a private pond for the embryo.*

***Q51. [SLO 4] Give an example of ectothermic and endothermic animals.***

*Ans: Ectotherm (poikilotherm): frog, lizard. Endotherm: birds and mammals.*

***Q52. [SLO 4] Name the vertebrate classes with two-chambered heart, four-chambered heart, air sacs and placenta.***

*Ans: Two-chambered: fishes. Four-chambered: birds, mammals (crocodile). Lungs with air sacs: birds. Placenta: eutherian mammals. Amniotic egg: reptiles, birds, mammals.*

***Q53. [SLO 4] Name the adaptations of birds for flight.***

*Ans: Feathers, wings, light bones with air spaces, air sacs, streamlined body, four-chambered heart and endothermy.*

***Q54. [SLO 4] What is the syrinx? What is the gizzard?***

*Ans: Syrinx: voice box of birds at the junction of trachea and bronchi. Gizzard: muscular part of the bird gut for crushing food.*

***Q55. [SLO 4] Name features found only in mammals.***

*Ans: Hair, mammary glands, diaphragm, three ear bones, pinna and non-nucleated RBCs.*

***Q56. [SLO 4] What is the difference between prototheria, metatheria and eutheria?***

*Ans: Prototheria lay eggs; metatheria give birth to immature young reared in a pouch; eutheria have a placenta and give birth to well-developed young.*

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***Q57. [SLO 4] What are monotremes? Why are they a link between reptiles and mammals?***

*Ans: Egg-laying mammals (platypus, echidna). They have mammary glands (mammal) and a cloaca (reptile).*

***Q58. [SLO 4] What are marsupials? Give examples.***

*Ans: Mammals whose young develop in the abdominal pouch (marsupium): kangaroo, koala, opossum.*

***Q59. [SLO 4] What is placenta?***

*Ans: An organ of exchange between maternal and fetal blood supplying nutrients and removing wastes.*

***Q60. [SLO 4] How do mammals differ from birds? Name shared adaptations.***

*Ans: Mammals have hair, mammary glands and give birth; birds have feathers and lay eggs. Shared: endothermy, four-chambered heart.*

***Q61. [SLO 4] Define homeothermic / endothermic and poikilothermic.***

*Ans: Endotherms keep a constant body temperature by metabolism (birds, mammals); poikilotherms (ectotherms) body temperature varies with surroundings (fish, amphibians, reptiles).*

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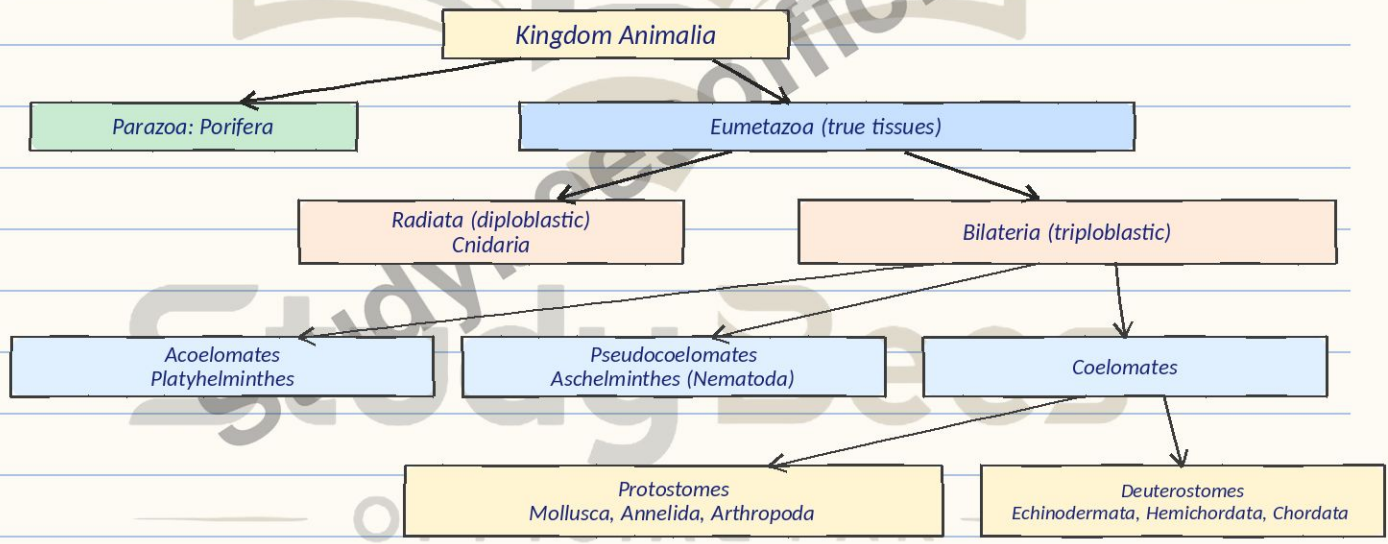
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# Chapter 10: Long Questions with Answers (SLO based)

## Q1. [SLO 1] Describe the criteria used for classification of animals.

**Ans:**

| Criterion | Groups |
|---|---|
| Tissues | Parazoa (sponges), Eumetazoa |
| Germ layers | Diploblastic (Cnidaria), triploblastic |
| Symmetry | Radiata, Bilateria |
| Body cavity | Acoelomate, pseudocoelomate, coelomate (protostome / deuterostome) |




*Protostome: blastopore becomes mouth; Deuterostome: blastopore becomes anus*

*Fig: Basis of animal classification (tissues, symmetry, coelom, blastopore)*

## Q2. [SLO 1] Differentiate between (a) diploblastic and triploblastic, (b) radial and bilateral symmetry, (c) acoelomate and pseudocoelomate.

**Ans:**

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| Pair | Difference |
|---|---|
| Diploblastic / triploblastic | Two germ layers with mesoglea / three germ layers |
| Radial / bilateral | Parts around central axis, sessile adaptation / mirror halves, motile adaptation |
| Acoelomate / pseudocoelomate | No body cavity, solid mesoderm / false cavity between mesoderm and endoderm |

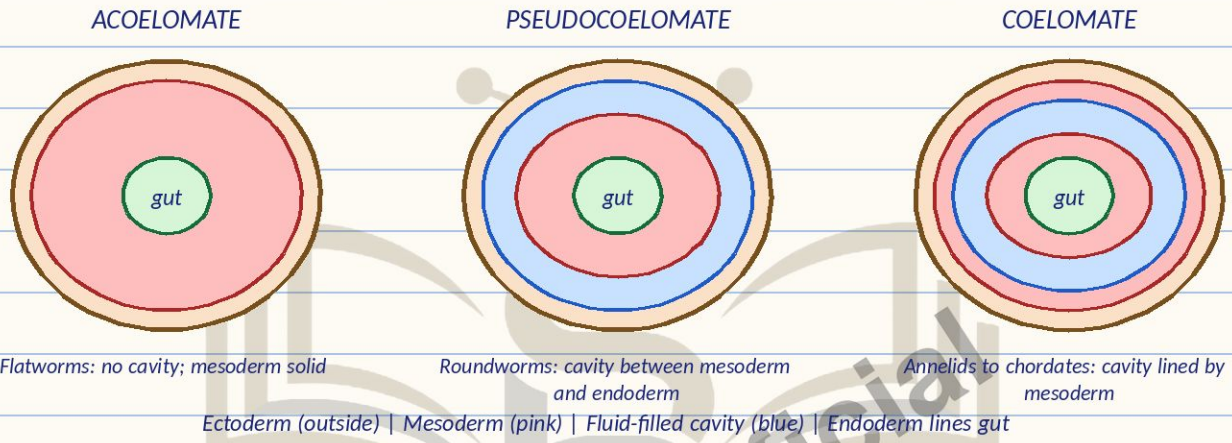



Fig: Acoelomate, pseudocoelomate and coelomate body plans (T.S.)

**Q3. [SLO 2] Describe the general characteristics, economic importance and examples of sponges and cnidarians.**

**Ans:**

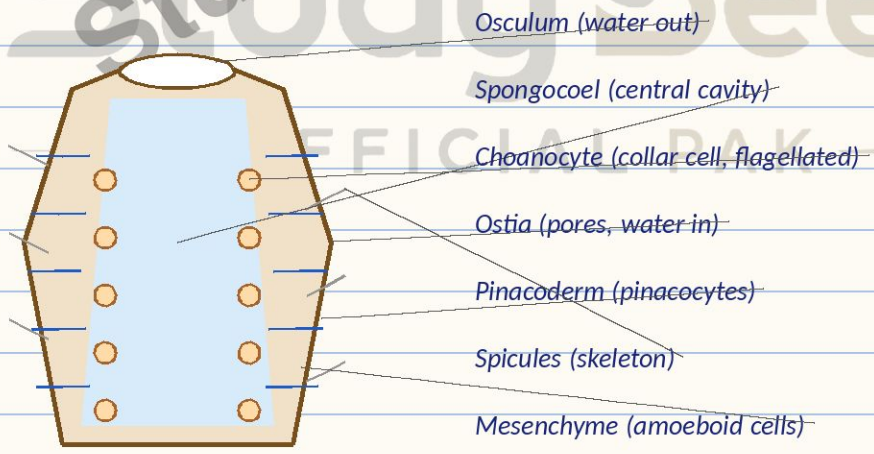



Fig: Structure of a sponge (Sycon type)

- Sponges: sessile aquatic; no tissues, no symmetry; pinacoderm, choanoderm, mesenchyme; ostia, spongocoel, osculum; spicules / spongin; budding,

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*gemmules; examples Spongilla, Sycon; uses: bathing, surgery, sound absorption.*

- *Cnidarians: aquatic; radial symmetry; diploblastic with mesoglea; gastrovascular cavity; nematocysts; polyps and medusae; examples Hydra, Obelia, jellyfish, coral; importance: reefs protect shores, jewellery, poisonous stings.*

***Q4. [SLO 2] Describe the general characteristics, economic importance and examples of Platyhelminthes and Aschelminthes.***

***Ans:***

- *Platyhelminthes: dorsiventrally flattened; triploblastic, bilateral, acoelomate; fission; monoecious; Planaria (free living), Fasciola, Taenia; harmful parasites (tapeworm, liver fluke, blood fluke).*
- *Aschelminthes: cylindrical, tapering, pseudocoelomate, dioecious; Ascaris, pinworm Enterobius vermicularis; parasites of man and animals.*

***Q5. [SLO 2] Describe the general characteristics, economic importance and examples of Molluscs and Annelids.***

***Ans:***

- *Molluscs: soft, unsegmented, coelomate; head, foot, visceral mass; mantle and shell; radula; trochophore; snail, slug, oyster, mussel, octopus; buttons, pearls, food; slugs and Teredo harmful.*
- *Annelids: metameric segmentation; true coelom as hydrostatic skeleton; setae, parapodia; closed circulation; ventral nerve cord; Nereis, Pheretima, Hirudo; fish food, soil improvement; leech parasite.*

***Q6. [SLO 2] Describe the general characteristics, economic importance and examples of Arthropods and Echinoderms.***

***Ans:***

- *Arthropods: jointed appendages; chitinous exoskeleton; haemocoel;*

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*metamorphosis; prawn, Cyclops, butterfly, spider; honey, wax, silk, pollination, food; pests and disease vectors.*

- *Echinoderms: marine; radial adults, bilateral larvae; spiny endoskeleton; water vascular system, tube feet; autotomy, regeneration; starfish, sea urchin, sea cucumber, brittle star; food, fertilizer, scavengers.*

## *Q7. [SLO 2] Describe the characteristics of chordates and their classification.*

***Ans:***

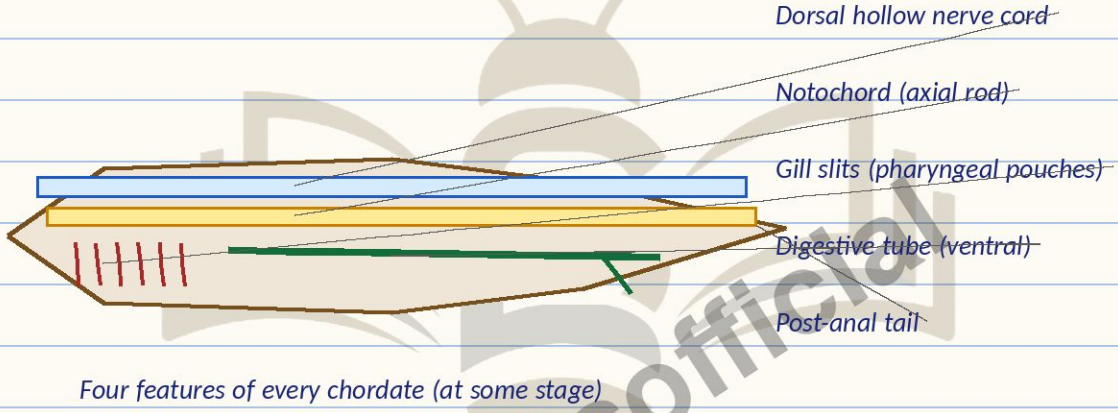



*Fig: Main features of chordates (generalized embryo)*

- *Notochord, dorsal hollow nerve cord, gill slits, post-anal tail.*

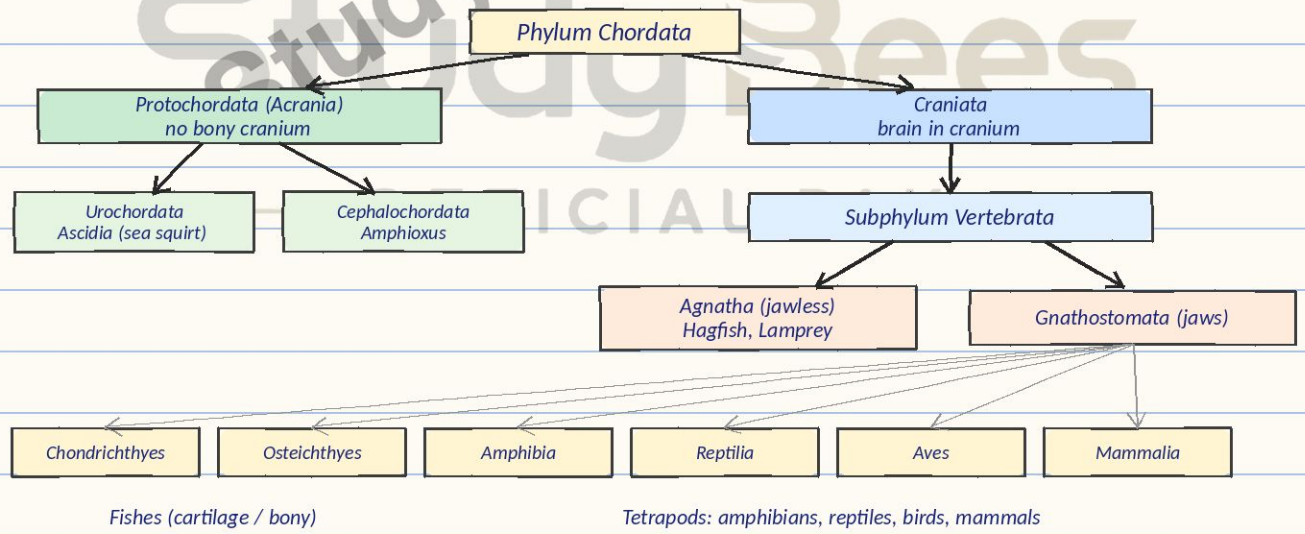



*Fig: Classification of phylum Chordata*

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***Q8. [SLO 3] Describe the general characteristics, evolutionary adaptations and examples of Chondrichthyes and Osteichthyes.***

***Ans:***

- *Chondrichthyes: cartilaginous skeleton; spindle body; ventral mouth; dogfish shark, skate. Osteichthyes: bony skeleton; streamlined; numerous vertebrae; swim bladder; Labeo rohita, sea horse.*
- *Lobe-finned fishes (Latimeria) gave rise to the first tetrapods, the amphibians.*

***Q9. [SLO 4] Describe the general characteristics, evolutionary adaptations and examples of Amphibia and Reptilia.***

***Ans:***

- *Amphibia: moist glandular skin, no scales; gills in larva, lungs and skin in adult; three-chambered heart; metamorphosis; poikilothermic; anamniotes; frog, toad, salamander. Not a very successful group due to dependence on water.*
- *Reptilia: dry cornified skin; incompletely four-chambered heart; internal fertilization; shelled amniotic eggs (amnion, allantois, chorion); poikilothermic; tortoise, lizard, snake, crocodile.*

***Q10. [SLO 4] Describe the general characteristics, evolutionary adaptations and examples of Aves and Mammalia.***

***Ans:***

- *Aves: feathers; wings; light air-spaced bones; beak; four-chambered heart; endothermic; air sacs; syrinx; gizzard; uric acid; ostrich, kiwi.*
- *Mammalia: hair; mammary glands; diaphragm; three ear bones; pinna; four-chambered heart; endothermic; deciduous and permanent teeth; viviparous.*

***Q11. [SLO 4] Show the heart structure in the vertebrate classes.***

***Ans:***

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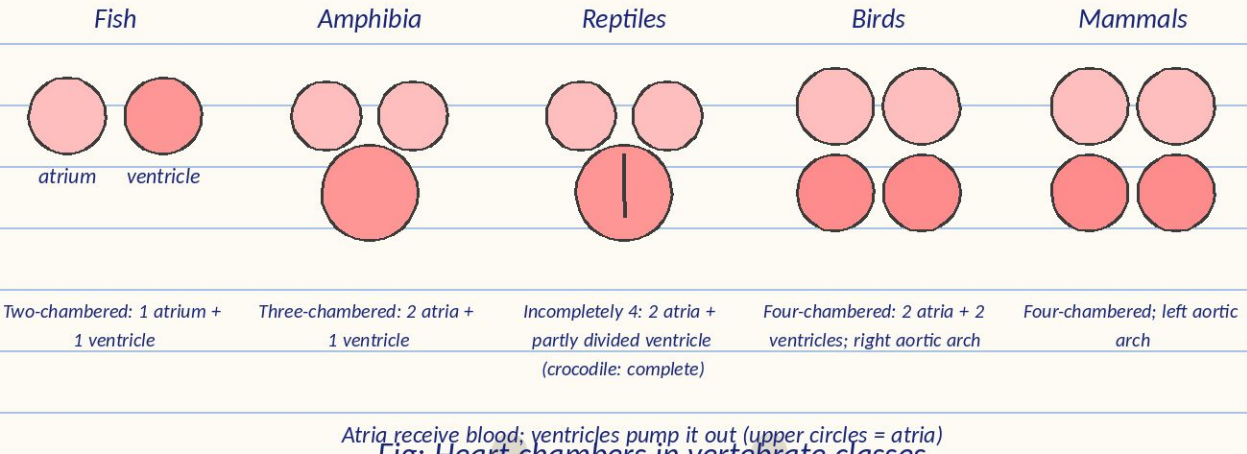



*Fig: Heart chambers in vertebrate classes*

- *Fish two-chambered; amphibians three; reptiles incompletely four (crocodile complete); birds and mammals four-chambered (right aorta in birds, left aortic arch in mammals).*

### *Q12. [SLO 4] Describe the characteristics and give examples of Prototheria, Metatheria and Eutheria.*

***Ans:***

| ***Subclass*** | ***Features / examples*** |
|---|---|
| *Prototheria* | *Egg laying; mammary glands + cloaca; Australia; platypus, echidna* |
| *Metatheria* | *Pouched; young immature; marsupium; kangaroo, koala, opossum* |
| *Eutheria* | *Placenta; young well developed; man, whale, elephant, bat, tiger* |

### *Q13. [SLO 4] How do mammals differ from birds? What adaptations do they share?*

***Ans:***

| ***Mammals*** | ***Birds*** |
|---|---|
| *Hair; mammary glands; viviparous (mostly)* | *Feathers; no mammary glands; oviparous* |
| *Diaphragm; teeth; three ear bones; left aortic arch* | *No diaphragm; no teeth (beak); one ear bone; right aortic arch* |
| *Lungs expandable; non-nucleated RBC* | *Lungs with air sacs; nucleated RBC* |

- *Shared: endothermy, four-chambered heart, amniotic egg ancestry, internal*

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*fertilization, dorsal hollow nerve cord, efficient respiration and excretion.*

***Q14. [SLO 4] Arthropods and vertebrates are highly successful groups on land. What shared characteristics are adaptive to land existence?***

***Ans:***

- *Waterproof covering (cuticle / keratinized skin); internal respiratory organs (tracheae / lungs); jointed appendages or limbs for support and movement; internal fertilization; protected eggs; efficient excretion; well developed sense organs and nervous system.*

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